Group 5

Ms. Kumar

Physics 140 Period 3

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Capstone Project Report



VIDEO LINK: https://www.youtube.com/watch?v=RVpRmz3-neM

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Project Overview

Criteria 1, 3, 11

Purpose

Given the increasing concerns regarding global warming, our project aims to solve one of the factors restricting us from approaching cleaner energy usage. Every day, we see people failing to dispose of their waste in the respective bins, whether it be throwing food-filled bags into recycling bins or recyclable cans into trash bins. Just in the past year, Americans have only a 32% recycling rate, leaving more than 2/3 of recyclable materials lost. Furthermore, not only are potential resources wasted, but 25% of recycling bins are contaminated, leading to the entire batch being unrecyclable. It is estimated that over 8 million metric tons of plastic enter the oceans annually, which is the equivalent of dumping in a garbage truck full of plastic every minute.

Our project aims to solve this problem in a microcosmic model of trash in the form of sorting & shooting yellow(trash) and white(recycle) ping pong balls into their respective bins. As the user of our product places their "waste" into our tube, it automatically gets sorted through the use of color sensors and launched through a motor-driven flywheel.

Addressing Physics Content

Our product addresses the specific topics required for this project by optimizing the amount of data we must collect vs. the concepts we can prove with each trial.

Kinematic Equations, Laws of Motion

1. Launch the ball with a known initial velocity and launch angle
2. Verify the formula we derived with laws of motion, kinematics, and drag

Drag, Terminal Velocity, Coefficient of Restitution, Energy Conservation, Turning Points

1. Drop a ping pong ball from 1 meter
    a. Use Tracker for final velocity vs calculated velocity for laminar flow rate
2. Use velocity before/after hitting ground for coefficient of restitution vs. manufacturer given value
3. Analyze Energy Conservation with bounce heights
4. Analyze Turning Points at peaks and collision points

CoM, MoI, Rotational Kinematics, Dynamics, Energies, Damped Harmonic Motion

1. Find COM of the turret & use a PID-loop to demonstrate Damped Harmonic Motion
2. Calculate turret's Moment of Inertia by using T = I*alpha with motor, compare with CAD

Existing Knowledge and Research

Criteria 2

Growing public concern for the efficacy of waste-reduction programs has led to many innovations in how we manage trash across the world, helping to alleviate pollution and increase reuse of materials. Below is a brief literature review we compiled of some existing projects described in research papers.

1. Automated Sorting Systems: Ability to sort types of materials(i.e. Ferrous vs nonferrous metals, plastic vs glass, etc)
    a. Sensors -- Electro-magnetic detection of metals & use of light to determine differences between translucent/opaque materials
    b. Control Systems -- Usually microcontrollers are used due to their compactness and the nature of trash sorting not necessitating massive computing capabilities
2. Social engineering through designing recycling bins
    a. Type of bins -- Studies found that multi compartment bins placed next to trash cans increased recycling %
    b. Signs -- The p-value of this did not indicate a significant causation, but combined with other factors there is a correlation between labels and recycling efficiency
    c. Public design -- Opening the design of recycling bins to the public increases participation & engagement, leading to a significant increase in interest
3. Innovative methods for sorting materials
    a. Frequency-based sorting -- using sound the object emits when hitting the ground OR emitting sound waves and classifying object based on vibrations
    b. Sensing -- Inductive proximity sensors are used to separate metals, LED and LDR are used together to enhance precision when sorting paper and plastic

Investigations

Criteria 3, 4, 5, 6

In each of the following investigations, we conduct a specific experiment to determine certain properties of various aspects of the project or verify derived formulas with real data.

We conducted the following investigations, not necessarily ordered in any way:

1. Projectile Motion and Drag
2. Terminal Velocity and Coefficient of Restitution
3. Controlling Initial Velocity
4. Project Trials / Testing the Formula
5. Center of Mass, Moment of Inertia (CAD Simulation)
6. Moment of Inertia (Turret), Rotational Physics, Damped Harmonic Motion

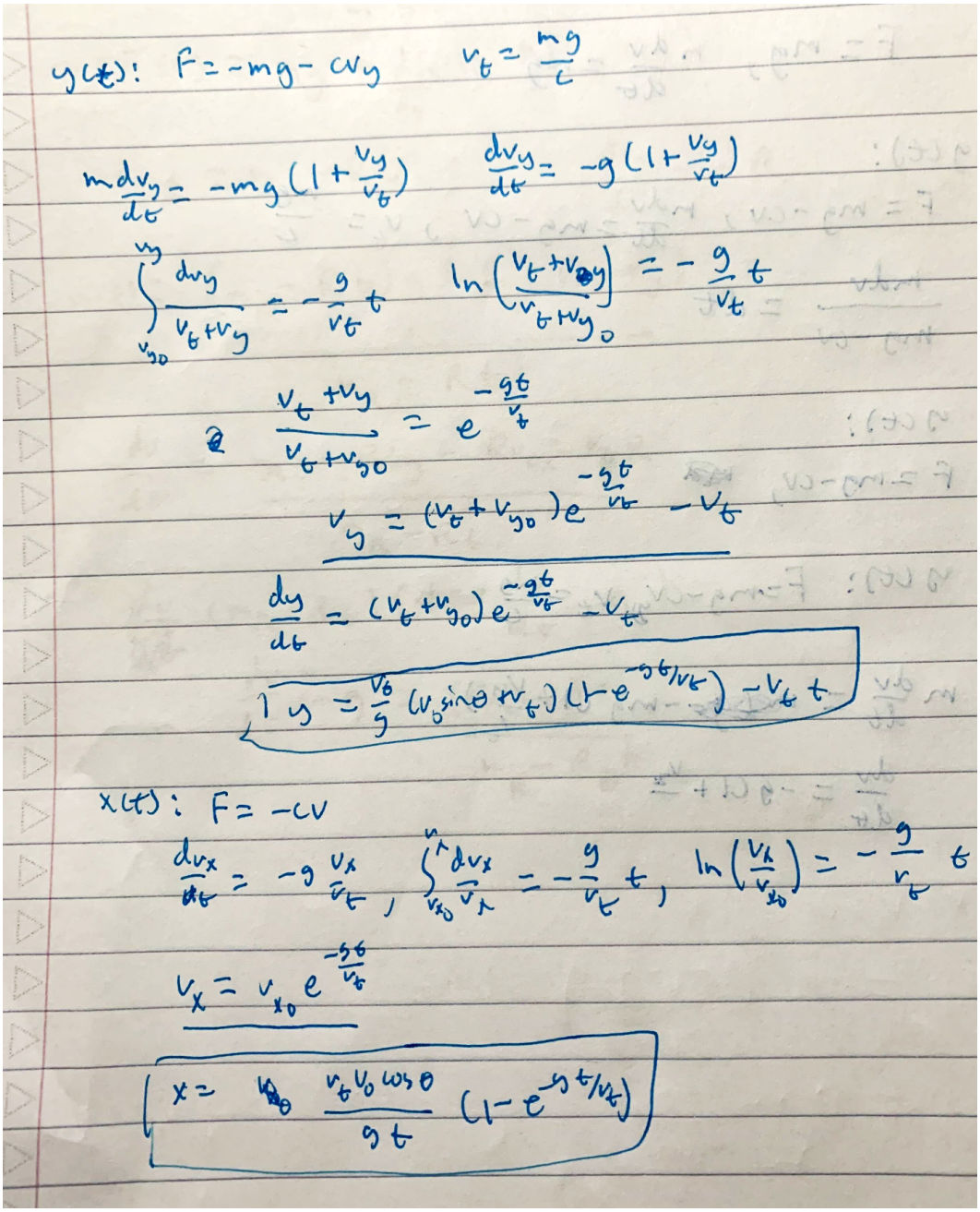
Projectile Motion and Drag

Purpose

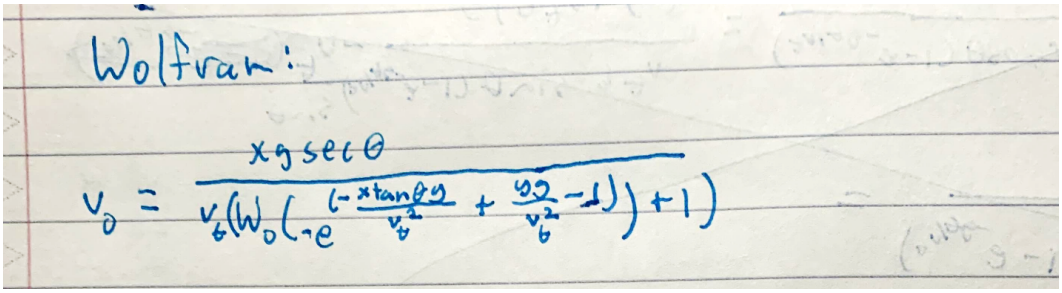
Using what we have learned about kinematics, projectile motion, drag, and terminal velocity, we will derive a hypothesized formula for the initial velocity required to launch the ping pong ball at a given target that is x meters horizontally and y meters vertically away. This formula would have the dual-purpose of also being used in the targeting process for our launcher.

Computational Model

We began with the force that the ball would experience in the horizontal and vertical directions and used **Newton's laws** and **kinematics** to derive x(t) and y(t).



Now that we have x(t) and y(t), we need to calculate a v_0 in which there exists a t such that x(t) = target x and y(t) = target y. We tried to manually derive further, but got stuck and instead used Wolfram Alpha to solve for us. We treated x, y, v_t, g, and θ as constants and asked Wolfram to solve for v_0.

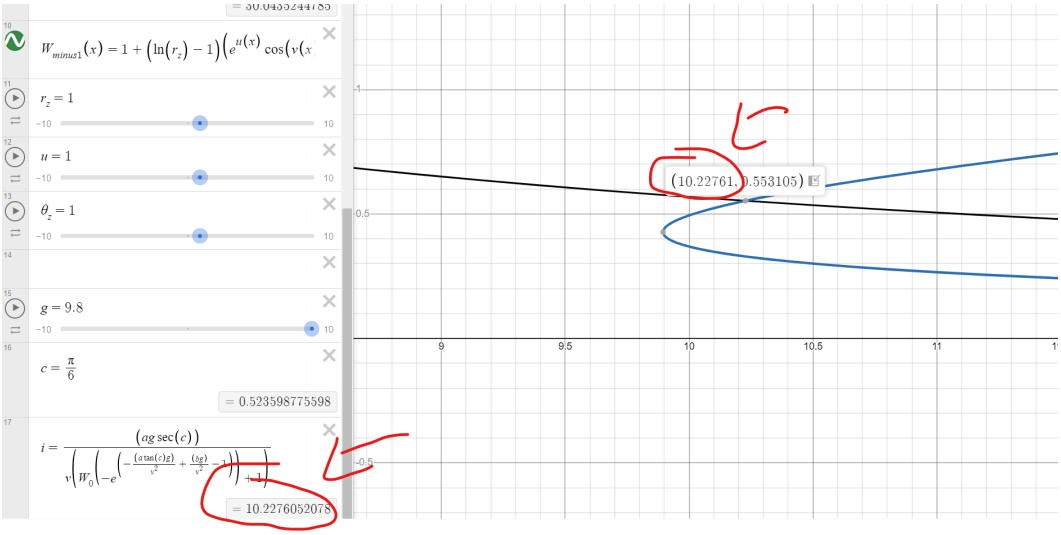


Conceptual Model

This formula makes sense because the initial velocity required increases as the x target increases. It also makes sense that gravity is in the numerator because a larger gravity would decrease the time of flight and require a larger initial velocity to cover the same distance.

Visual Model

We verified that this is true by graphing on desmos and finding that the intersection of our y(t) and x(t) formulas when we graph it with v_0 on the x-axis and t on the y-axis intersected at the same v_0 value as this formula provided.



However in order to use this formula we must first calculate some physical constants, the terminal velocity and the conversion between target motor velocity and observed ping pong ball velocity.

Terminal Velocity and Coefficient of Restitution

Purpose

To use the above projectile motion formula, we first need the terminal velocity of the ball. When the ping pong ball hits the ground, it would bounce back and this would also be a convenient time to calculate the coefficient of restitution, analyze conservation of energy, and turning points.



Materials

Meter Stick, Camera, Ping Pong Ball

Procedure

1. Set the meter stick up vertically, place the camera perpendicular to the plane of the meterstick and with both the ping pong ball and ground visible.
2. Hold the ping pong ball at the top of the meter stick
3. Start recording and drop the ping pong ball
4. Repeat 10 times to increase precision

Data

(We initially did it 5 times, but redid it more accurately a second time 10 times for the terminal velocity investigation)

	V_y Before(m/s)	V_y After(m/s)	Coefficient of Restitution (e)	Initial Height	Bounce Height
Trial 1	-3.002	2.713	0.9037308461	1.06	0.66
Trial 2	-3.119	2.797	0.8967617826	1.01	0.64
Trial 3	-3.114	2.798	0.8985228003	0.99	0.62
Trial 4	-3.102	2.721	0.8771760155	1.02	0.63
Trial 5	-3.092	2.696	0.871927555	0.99	0.62
Average	**-3.0858**	**2.745**	**0.8895586234**	**1.014**	**0.634**
STDEV of e:	**0.01411786261**				

The fall time for all 10 slow motion trials:

Trial #	Fall Time (sec)
1	0.479
2	0.475
3	0.466
4	0.488
5	0.471
6	0.479
7	0.483
8	0.496
9	0.475
10	0.475

Computational Model

$$y(t) = \frac{v_t}{g}\left(v_0 \sin\theta + v_t\right)\left(1 - e^{-\frac{gt}{v_t}}\right) - v_t t$$

Taking our y(t) formula from before, we know all the variables other than v_t allowing us to solve for v_t. Solving this equation is quite difficult, but it will only need to be calculated once so we can numerically optimize it programmatically rather than getting a formula.

We then took the average time of falling over 10 runs which was: 0.480s

Here is the code we wrote:

```python
import numpy as np
from scipy.optimize import root_scalar
import matplotlib.pyplot as plt

# Define the equation to solve
def equation(v):
    g = 9.81  # gravitational acceleration in m/s^2
    m = 0.0027  # mass of the object in kg
    t = 0.48
      # time it takes to fall 1 meter
    y_target = -1.00  # target height (1.02 meters)

    # Position function y(t)
    y_t = v * v / g * (1 - np.e ** (-g * t / v)) - v * t

    # Return the difference from the target height
    return y_t - y_target

# Check the signs at the endpoints of the bracket
a, b = 10, 1000
print(f"equation({a}) = {equation(a)}")
print(f"equation({b}) = {equation(b)}")

# Use root_scalar to find the root within a reasonable range
# If the signs are different, we can use the bracket for root-finding
if equation(a) * equation(b) < 0:
    result = root_scalar(equation, method='brentq', bracket=[a, b])

    # Print the entire result to inspect the structure
    print("Root Scalar Result:", result)

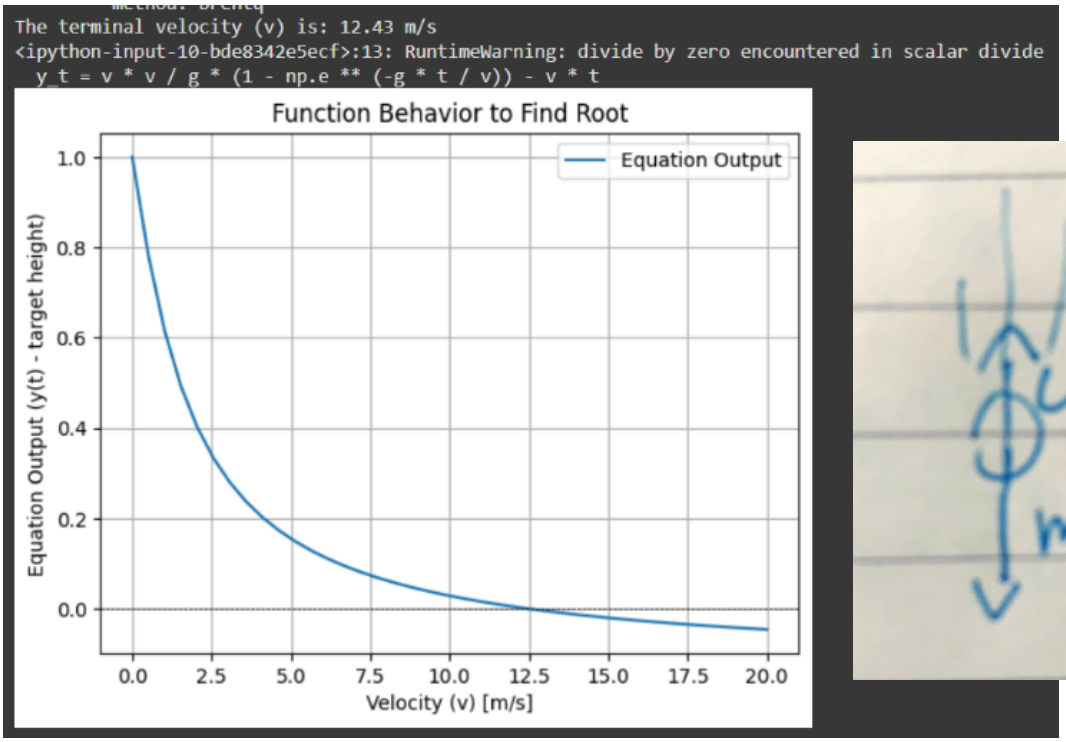
    # Check if result has 'root' attribute
    if hasattr(result, 'root'):
        print(f"The terminal velocity (v) is: {result.root:.2f} m/s")
    else:
        print("Solution not found or result structure is different.")
else:
    print("The function does not change signs between 10 and 100, try a different bracket.")

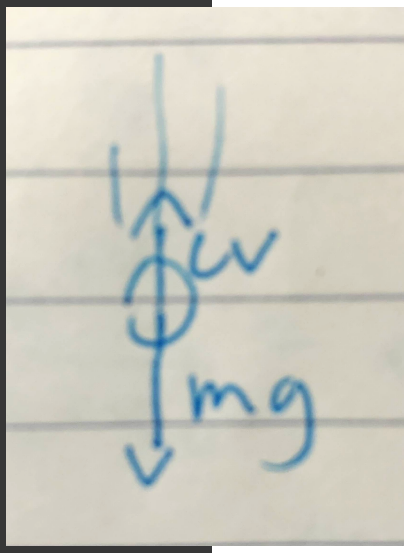
# Optional: Plot the function to visualize the root
v_values = np.linspace(0, 20, 40)
y_values = [equation(v) for v in v_values]

plt.plot(v_values, y_values, label="Equation Output")
plt.axhline(0, color='black', linewidth=0.5, linestyle='--')
plt.title("Function Behavior to Find Root")
plt.xlabel("Velocity (v) [m/s]")
plt.ylabel("Equation Output (y(t) - target height)")
plt.grid(True)
plt.legend()
plt.show()
```

Running this returned a **terminal velocity of 12.43m/s**. However just decreasing the average time by .01s would alter this number significantly to 19m/s, so we can only use this number as an estimate.

Visual Model





Conceptual Model

This makes sense because the drag will slow down the process of falling and decreasing the average falling time should increase the terminal velocity, which is inversely proportional to drag.

Computational Model

The coefficient of restitution of the ping pong ball against the ground can be determined by recording the speed of the ball relative to the ground before and after the ball collides with the ground.

$$e = \frac{V_{fb} - V_{fa}}{V_{ia} - V_{ib}}$$

Ex: Trial 1: Speed approach = 3.002, Speed separation = 2.713, e = 2.713/3.002 = .904

The average coefficient of restitution we calculated was .89 with a standard deviation of .014

This lines up with the official coefficient of restitution of ping pong balls of 0.89-0.92 (from TuHS Physics).

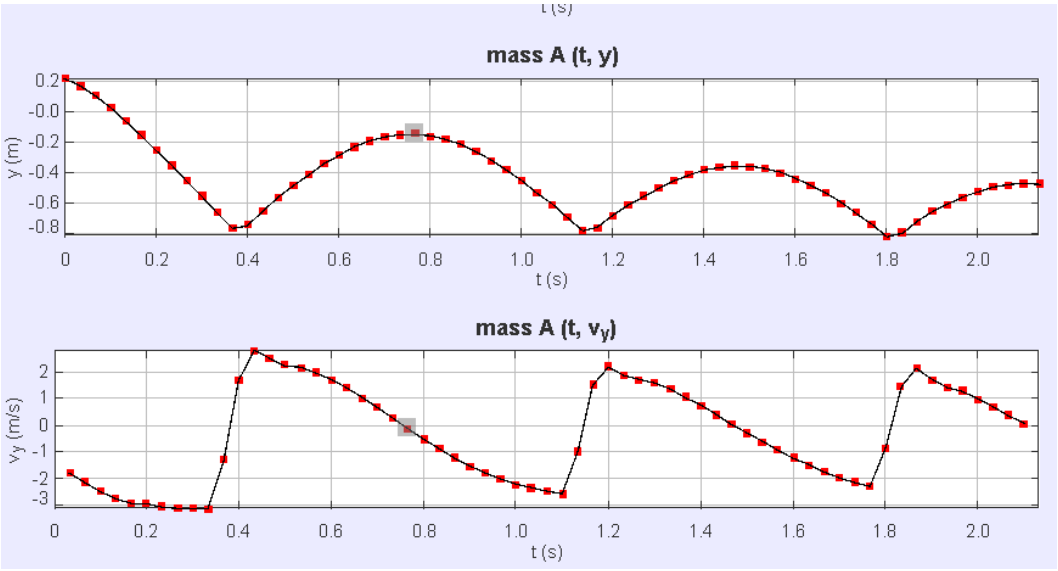
Since the ball is dropped from an average height of 1.014m and bounces back to an average of 0.634m, the energy of the system changes by an average of .0027kg*9.81m/s^2*(.634m-1.014m) = -.01J

Since the total mechanical energy decreased through this system, the **total mechanical energy was not conserved**.

Turning points occur when the velocity is 0. This occurs at two different points, the peak of each bounce as well as the moment when the ball collides with the ground. This is because ME = KE + PE, at the peak of the bounce, ME = PE, so KE = 0. KE = ½ m * v^2, so v =0.
At the collision between the ball and the ground, the ball accelerates from a negative velocity to a positive velocity, so it must have 0 velocity at a certain point in time.

Visual Model

The graph of y velocity intersects at zero at the peaks of bounces and collision points.



Conceptual Model

Since the ball bounces to a lower height, it makes sense that the coefficient of restitution is less than 1 because the velocity that the ball bounced back with relative to the ground was lower than the velocity that the ball fell towards the ground with.

Additionally, it is expected that **energy is not conserved** because drag and inelastic collisions are both non-conservative forces that dissipate energy.

The **turning points** are at the peaks of the bounces and the ground because those are the points where the ball changes direction by having zero velocity and acceleration in the opposite direction.

Conclusion

By recording the time it took for the ping pong ball to fall 1 meter, and numerically calculating the terminal velocity that would result in such a time, we were able to determine the

terminal velocity of a ping pong ball to be about 12.43 m/s. Our model is reasonable because as the time it takes the ball to fall decreases, the calculated terminal velocity increases. We were also able to calculate the coefficient of restitution to be 0.89, which means that energy is not conserved. This is expected because both drag and inelastic collisions are non-conservative forces.

Controlling Initial Velocity

In order to use our formula, we would also need to be able to control the initial velocity accurately.

Purpose

The purpose of this investigation is to model the relationship between the target velocity of the outer edge of the wheel and the observed initial velocity of the ping pong ball, so that we can accurately control the initial velocity of the ping pong ball to be used in the formula.

Materials

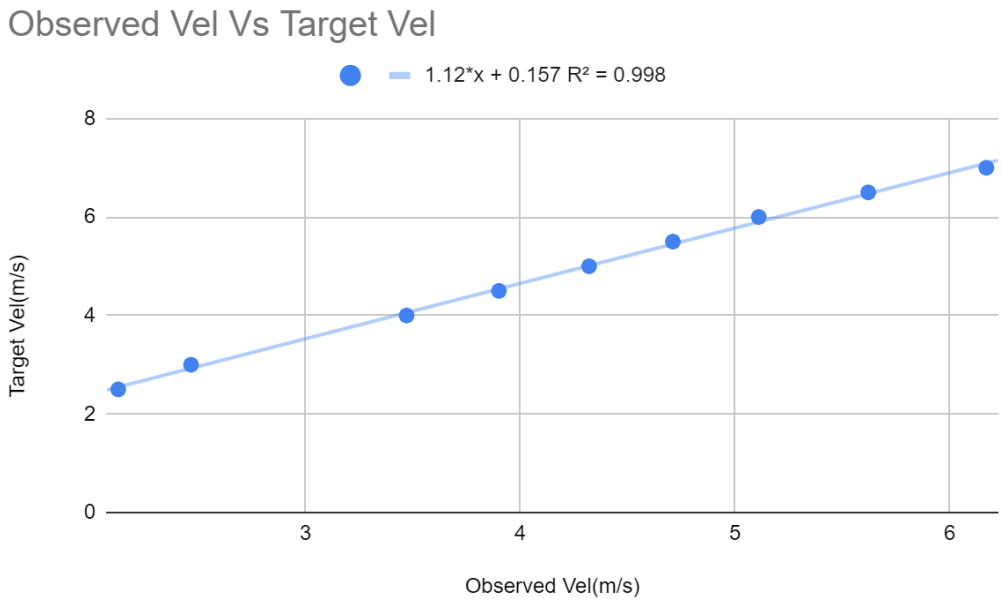
Meter Stick, Camera, Launcher, Ping Pong Ball

Procedure

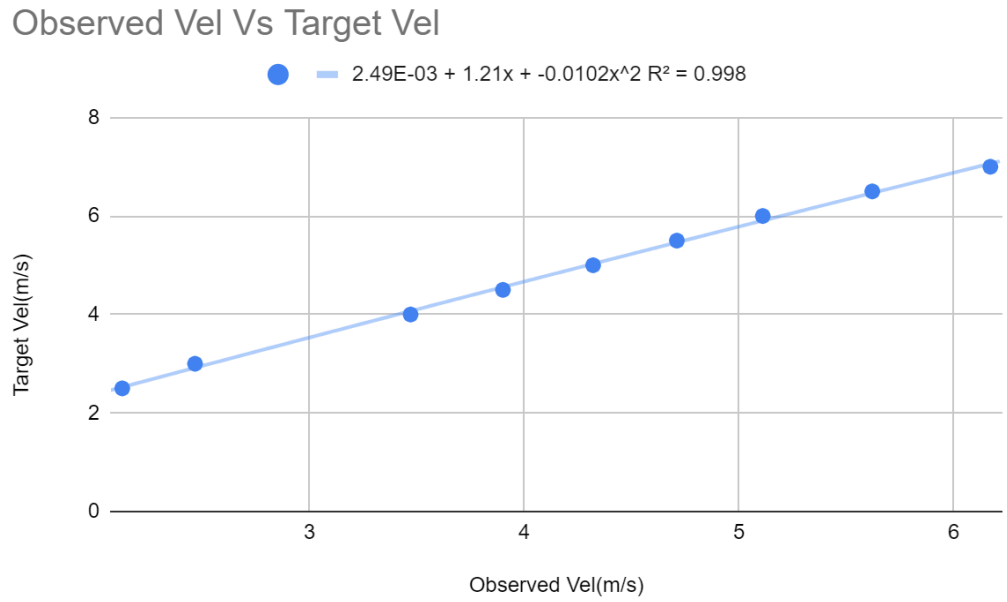
1. Place the launcher on the ground, place the meter stick in front of the launcher in the direction it is shooting
2. Place the camera perpendicular to the plane of shooting and the meter stick
3. Set the target velocity in m/s for the motor to spin the surface of the wheel at times 2
4. Start recording in slow motion and feed a ping pong ball into the launcher
5. Repeat for every velocity from 2.5 m/s to 7.5 m/s

Data

Linear Model



Quadratic Model



Conclusion

The relationship between the target motor velocity and the observed initial velocity is quadratic, as despite having identical R^2 values of .998, the b value of the quadratic model is nearly 0, which is not the case for the linear model.Product Trials / Testing the Formula

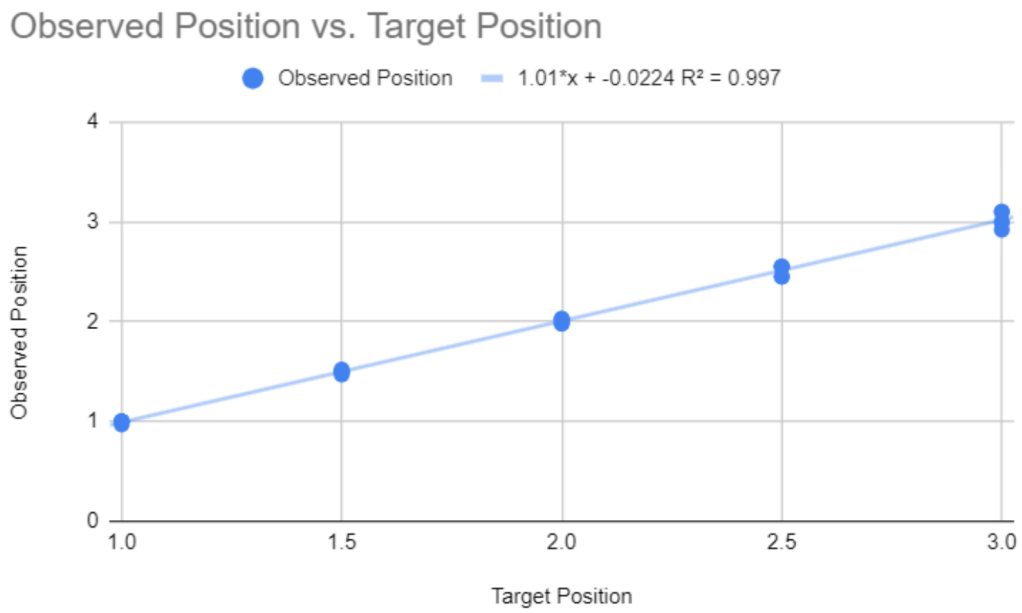
Materials

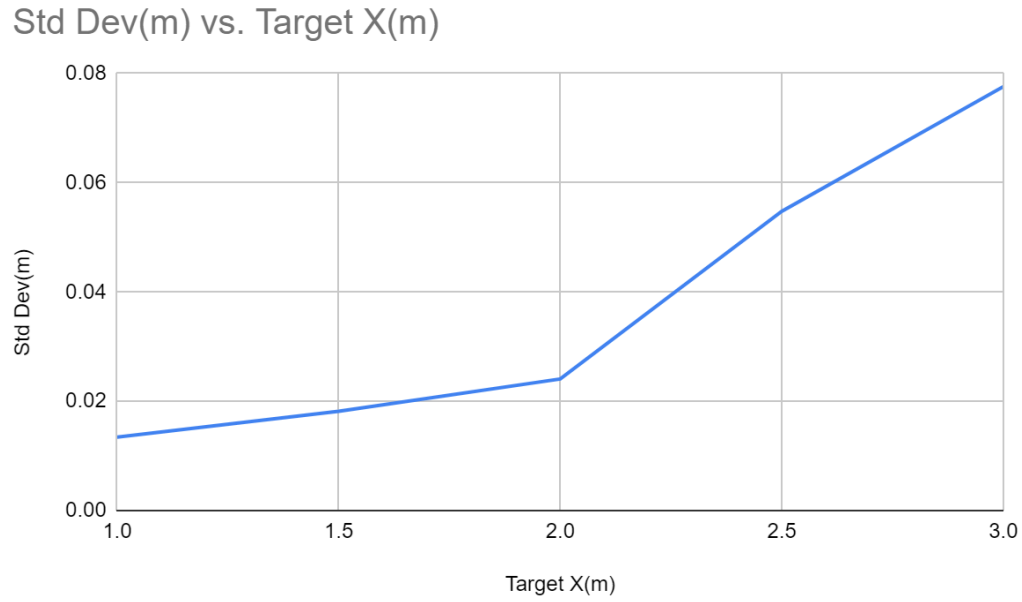
Elevated Surface, Measuring Tape, Launcher, Ping Pong Balls, Camera

Procedure

1. Place the launcher on an elevated surface
2. Measure the height of the elevated surface
3. Place the measuring tape on the ground with the 0 at the launch point
4. Place the first target(red) on your target x position
5. Start the program and wait for first target scan to finish
6. In the 2 seconds before the next scan starts, swap out the first target for the second target (blue)
7. Wait for the second target scan to finish
8. Remove the second target record the measuring tape at the target position before the launcher starts shooting
9. Wait for the launcher to shoot 5 balls at the target
10. Record the landing position of each ball
11. Repeat at 5 different positions from 1m to 3 m

Data





Ball In Box Success Rate: 21/25

Qualitative Observations

The white balls almost always shot shorter than the orange balls, forming two clumps of data for each distance.

Product Reliability Analysis

The R^2 value of our trendline is .997, indicating that the formula is a good fit for our data. The slope is 1.01, which is very close to the ideal of 1 and it also has a very small b value as expected. However, as the target moves further away, the precision of our shooter decreases. From 1 m to 2m, our launcher is quite precise, with a less than 3 cm standard deviation, but from 2.5m and beyond, the inconsistency increases drastically and becomes much less reliable. We had an **overall success rate of 21/25, 84%**, with 3 of the failures occurring at the 3 meter target and 1 at the 2.5 meter target. 3 of the failures were white balls going too short, verifying our observation that we lose consistency past 2.5m and that white balls don't shoot as far.

Assumptions, Limitations, & Error Analysis

Some possible sources of the increased variability are:

1. Different colored ping pong balls were not launching the same way
    a. We noticed that the white balls constantly fell shorter than the orange balls
        i. We tuned the motor velocity to ball velocity model using orange balls, so making the assumption that different colored balls would behave the same way was incorrect
2. Inconsistent launch angle due to design
    a. We noticed in Tracker that the launch angle varied from 23 degrees to 28 degrees with an average of 25 degrees, when our target was 30 degrees. We launched these at 25 degrees in the code to minimize error due to this variability
    b. This was likely due to our design launching the ball too late by having a circular track that was longer than anticipated and unpredictable contact points in the track
        i. This could have been avoided by designing the launcher with a flat incline to avoid having such a track
3. Inaccurate terminal velocity calculations due to procedure
    a. We had to increase the terminal velocity from the calculated 12.43 to 12.8 in order to accurately hit the far range targets, indicating that we still may not have an accurate terminal velocity
        i. Dropping the ball from a higher point would have given us a higher resolution when calculating the terminal velocity
4. Inaccurate motor velocity → ball velocity conversion model due to design
    a. We don't have a really good explanation for why the model would be quadratic, so we are not confident in the accuracy of this model
    b. There is also large variability in the initial velocity of the ping pong ball even when the wheel is spinning at the same speed
        i. This could potentially be addressed by using a two wheel design which wouldn't have the unpredictable friction of the ball against the track.
5. Very slow and slightly inaccurate scanning

a. The camera had significant motion blur, so we would have to wait for the turret to settle before scanning
    i. This could be addressed by purchasing a global shutter camera, which wouldn't have this motion blur issue
b. The camera we used to calculate the position of the targets was low resolution (640x480) and we had to tune its distortion coefficients for accurate position calculation
    i. Using a higher resolution camera would increase the resolution and accuracy of calculated positions
    ii. Using a no distortion camera or a camera with well known distortion coefficients could eliminate the uncertainty from tuning the distortion coefficients

Formula Conclusion

C: We claim that given a target x and y distance, the following formula can be used to calculate the required terminal velocity to hit the target:

$$v_0 = \frac{(xg \sec\theta)}{v_t\left(W_0\left(-e^{\left(-\frac{x\tan\theta g}{v_t^2}+\frac{yg}{v_t^2}-1\right)}\right)+1\right)}$$

E: (in addition to the CCV evidence presented on page 6)

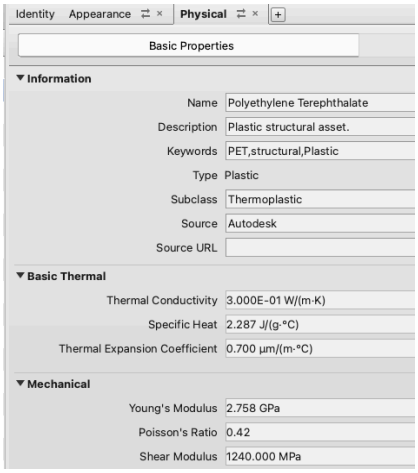
The R^2 value of our trendline is .997 with a b-value .0075% of the maximum value.

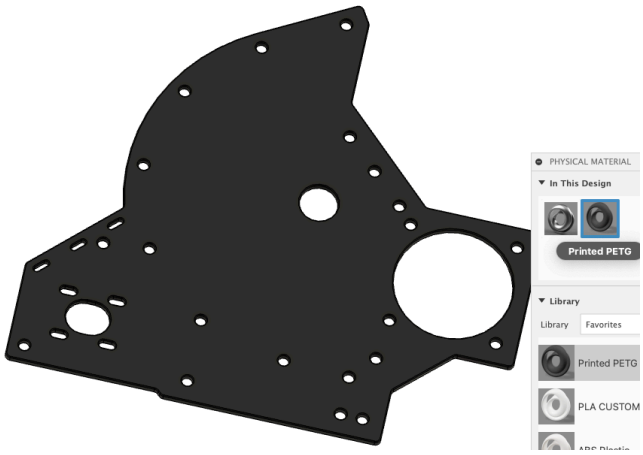
R: We derived the formula starting from Newton's laws and kinematics and visually cross checked the only step that was out of our control, wolfram alpha, by plotting out the intersection of our derived formulas for x(t) and y(t) and verifying that they had the same initial velocity. Additionally, the direction of increase/decrease that the x and g constants in the formula on the target initial velocity makes conceptual sense. Lastly, when we shot our launcher with the initial velocity specified in the formula, we were able to accurately hit the targets on average, with an R^2 value > .99 and a b-value < 1% of the maximum y-value, indicating that our model is accurate.

Center of Mass, Moment of Inertia (CAD Simulation)

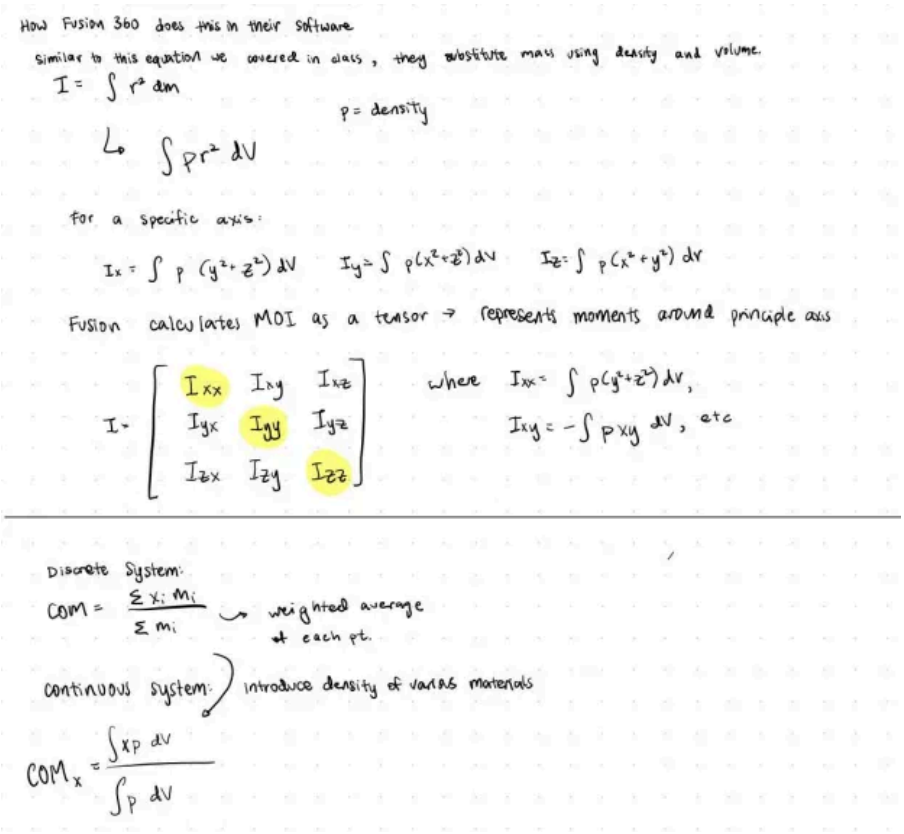
Procedure

1. Weigh each component and use volume calculated by Fusion 360 to calculate density of each material used (e.g. PETG, PLA, Acrylic, Compliant Wheel, COTS, etc) and create custom material profile

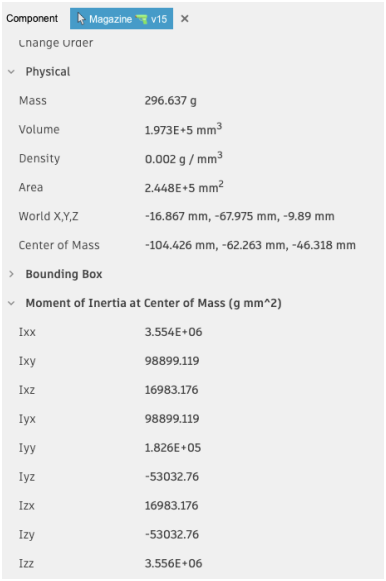




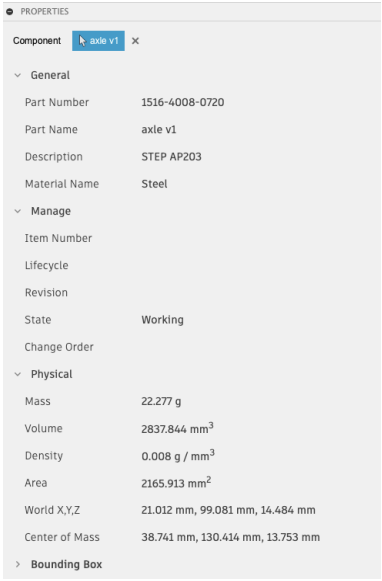
2. Assign physical materials to all pieces of the shooter and magazine (rotating components).
3. Determine the center of mass of the shooter and magazine using Fusion 360 properties
4. Determine the moment of inertia from the center of mass of the shooter and magazine using Fusion 360 properties.

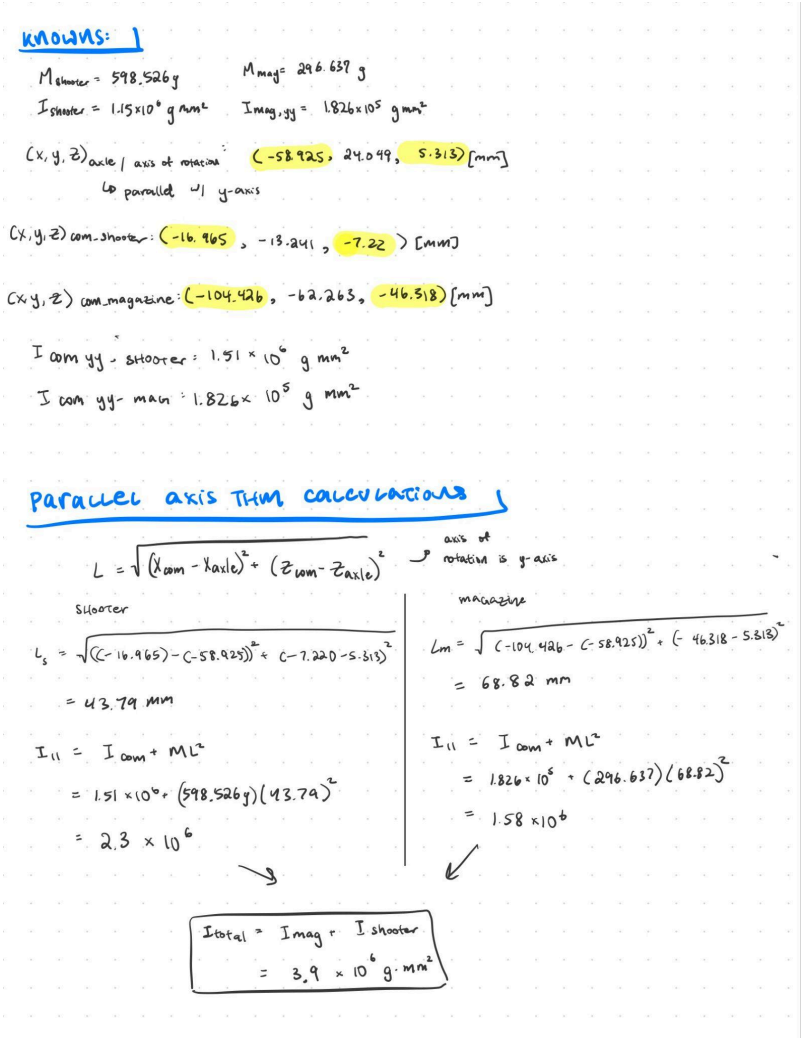


5. Determine the distance from the center of mass of the shooter and magazine from the axis of rotation
6. Use the parallel axis theorem to calculate the moment of inertia using their respective COM and MOIs. Add together the MOI of the shooter and magazine to get the total moment of inertia about the pivot.









Conclusion / Data Analysis

This calculation of 3.9 * 10^6 g mm^2 for the moment of inertia about the pivot is a full order of magnitude greater than the analysis on the data collected by the actual turret reports as seen in the section below. Potential sources of error could come from variability in the density of the material or the accuracy of the real life assembly to the CAD (exclusion/inclusion of minor components like the axles, clamps, pulleys). The infill of the 3D printed pieces was not accounted for exactly in our calculations within Fusion 360, and there are almost certainly other inaccuracies with other materials used in the CAD that do not reflect reality. Another probable source of error is with the data collection source itself. The data from the turret was collected via an encoder on the motor, and it's possible this encoder provided faulty data as evidenced by the data itself, which repeatedly reported repeat values when that shouldn't have been possible. However, the likeliest, greatest source of error was the fact that we forgot to account for any internal motor resistance. Unaccounted resistance in the gearbox of the motor would've changed the physics of how the turret actually behaved in real life, and this may not have reflected what the motor encoder reported by a large amount.

Moment of Inertia (Turret), Rotational Physics, Damped Harmonic Motion

Procedure

1. Initialize the program
2. Displace the turret 90 degrees from the center
3. Using the program apply a resistive torque proportional to the opposite of the current displacement
4. Have the program log position, velocity, and torque

Data

https://tinyurl.com/inertiadata

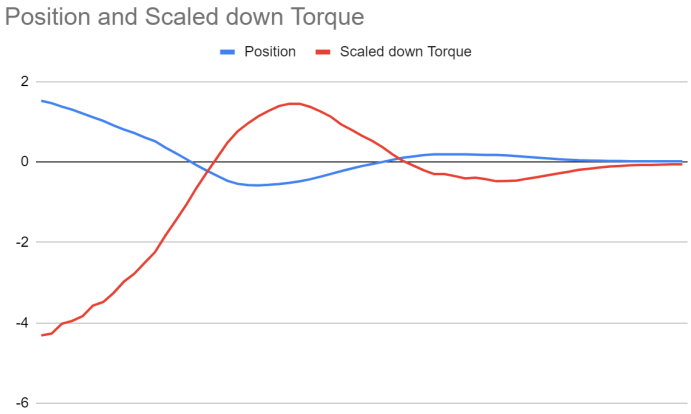
Computational Model

The data shows torque values collected by the motor encoder as the turret moves. We adjusted the torque values because there was some torque recorded while velocity and position were 0 (accounted for by static friction force), and then plotted torque against time. We then took a midpoint Riemann sum as an approximation of the total change in angular momentum. Dividing the sum of that by the total change in velocity (the data recorded was a deceleration to 0), we were able to derive the moment of inertia of the turret in reality. This value was .0204 kg m^2, an order of magnitude lower than as predicted by Fusion 360 simulation. Accounting for this error is done in the previous section.

Conceptual Model

The data shows torque values collected by the motor encoder as the turret moves. We also collected rotational velocity data, which we used to extrapolate acceleration values. It can be shown when plotting rotational acceleration against torque, that the two share an approximately linear relationship, where the slope is approximately the moment of inertia of the turret as calculated using data values below. This serves as evidence for the concept torque = I*a. Also, when taking two close times, the data values associated with those times approximately demonstrate the rotational kinematic equations at an instant. Obviously, as the data is not at an instant and the rotational acceleration is not constant, the data will not reflect exactly, but they are relatively close. This serves as evidence for the rotational kinematic equations.

Visual Model



Computational Model

Max Velocity: 17 rad/s

Turning points: (1.52 rad, 9.11s), (-.58 rad, 9.27s), (.189 rad,9.45s)

The **maximum rotational energy** of the turret in accordance with the formula on the right is:
$0.5*17s^{-1}*17s^{-1}*.0039kgm^2 = .563J$

$$K = \frac{1}{2} I\omega^2$$

To calculate the **damping constant** of our approximate simple harmonic motion we solve for γ using the formula $\gamma = -1/\Delta t * \ln(A_n/A_{n+1})$. Applying this to our first interval yields:

$\Delta t_1 = 9.27\text{-}9.11 = .16s$, $A_0 = 1.52$, $A_1 = .58$
$\gamma_1 = 6.012\ s^{-1}$
Second Interval:
$\Delta t_2 = 9.45\text{-}9.27 = .18s$, $A_0 = .58$, $A_1 = .189$
$\gamma_2 = 6.244\ s^{-1}$
Averaging these yields a **damping constant of approximately $6.128s^{-1}$**

Conceptual Model

This nonzero damping constant makes sense because we observed the harmonic motion dampen and eventually come to a stop.

Conclusion

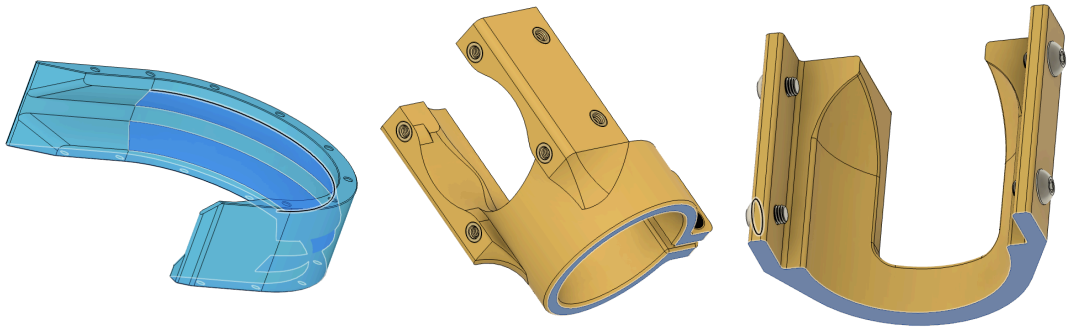
Our program could not effectively control the torque to be exactly proportional to the opposite of displacement, so the turret behaved only as an approximation of simple harmonic motion, but despite this, by recording the displacement and time at the turning points we can approximate the damping constant of these oscillations.

Material Choice

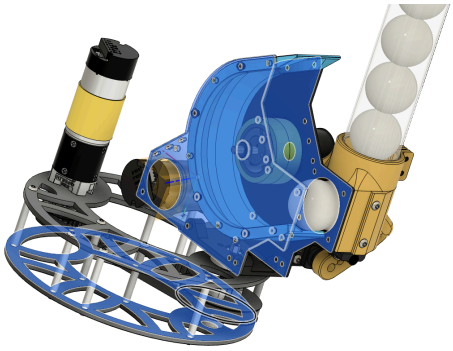
Criteria 9

When deciding to choose materials, we took into 4 considerations.
C1. Ease of Manufacturing
C2. Robustness
C3. Availability of tools
C4. Cost

3D printed parts (C1)



These parts are specifically designed to be 3D printed, created with a flat face and chamfers/fillets to allow it to be printed without support structures (C1). 3D printing allows use of complex geometries (C1), making them easy to manufacture while still being relatively strong for our needs (C2).



These highlighted parts are 2D drawings extruded by 3mm - 4mm thickness, allowing it to be manufactured out of a single sheet of material through fabrication methods like laser cutting. We ultimately decided to 3D print it as it was the tool we had on hand (C3).

Reason for Purchased / COTS parts

- Electronics
    - Color Sensor
    - Servo Motor
    - DC Motor
    - Control Hub
- Inefficient use of 3D printed material (convenience)
    - Ping Pong Balls
    - Compliant Wheel
    - Acrylic Tube
- Needed for Robustness
    - Standoffs
    - Axles, Clamps, Belts

Installation Manual

Criteria 10

Materials

- REV Expansion Hub
- Launcher
- 2x Bullet to JST VH wires
- 2x JST PH to JST XH wires
- 1x JST PH to JST PH wire
- USB hub
- Android Phone with installed apps
- Battery
- Laptop

Installation steps

1. The wires should come pre-plugged but if not, plug in as follows
    a. Turret motor and encoder plugged into port 2 on the expansion hub
    b. Flywheel motor and encoder plugged into port 1 on the expansion hub
    c. Color sensor connected to I2C bus 2
2. Once wired, place the hub to the left of the launcher when looking from behind the launcher, give the wires space so that they won't get caught when the turret rotates
3. Plug the expansion hub into the battery with the anderson power pole cable
4. Plug the usb of the expansion hub and into the usb hub
5. Power on the phone and open the FTC Robot Controller app
6. Plug the phone into the usb hub
7. Once the expansion hub light flashes green, plug the camera into the usb hub
8. Connect the laptop to the phone's wifi-direct network, the passkey can be seen by pressing the three dots on the phone app and selecting program and manage
9. Open a browser and go to http://192.168.49.1:8080/dash
    a. Now you are all set up!

Usage Manual

Criteria 10

Usage steps

1. Load the launcher with as many ping pong balls as you like
2. Point the launcher at the target, scanning works at +- 55 degrees, so you don't have to be precise
3. Connect the laptop to the phone's wifi-direct network, the passkey can be seen by pressing the three dots on the phone app and selecting program and manage
4. Open a browser and go to http://192.168.49.1:8080/dash
5. Select the program drop down on the top left and select the "Capstone" program.
6. Tape the paper targets to your desired targets, make sure the targets are facing the camera
7. Run the program

Works Cited

Criteria 13

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